

VIKING LONGSHIP

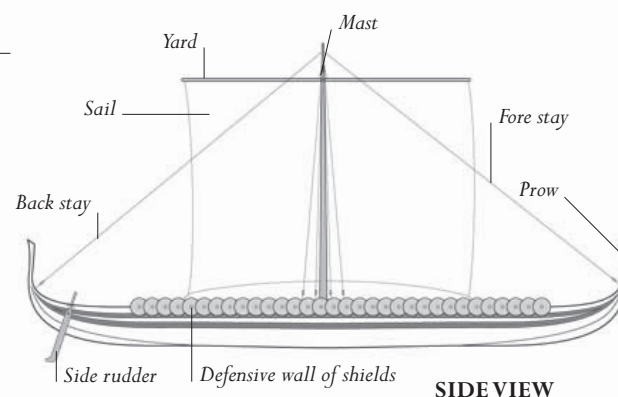
HAVHINGSTEN FRA
GLEN DALOUGH

The Viking longship was a swift, sturdy, and versatile naval craft. Propelled either by a sail or by oars, it was one of the fastest vessels of its era, and able to travel great distances on the open sea.

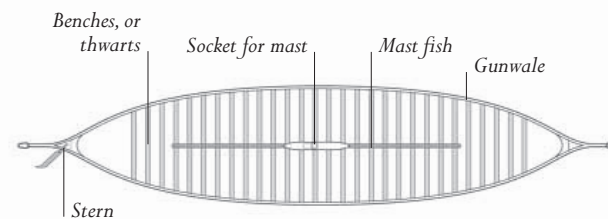
A longship's shallow draught enabled it to penetrate far upriver or beach itself at speed in an early form of amphibious assault. It could also be hauled over narrow necks of land between waterways. The warriors it carried mostly took part in land raids, but occasionally crude sea battles took place involving exchanges of missiles and boarding with hand weapons.

The longship shown here, *Havhingsten fra Glendalough* (Sea Stallion from Glendalough), is a reconstruction of a vessel excavated from Roskilde Fjord, Denmark, that was built around 1042. As far as possible, the

reconstruction used the techniques, tools, and materials of the Viking age, including timber from 300 oak trees. Steered by a side rudder, the longship has a strong keel and a high, curving prow and stern. It is clinker-built, meaning that its hull is made of overlapping planks, or "strakes", held together with iron nails. About 60 oarsmen – who doubled up as warriors on raids – would have rowed such a ship. Their muscle power could have delivered a steady speed of 5–6 knots, but with its sail raised and a favourable wind, a longship could probably have made up to 17 knots.



SIDE VIEW



PLAN VIEW

▲ LONGSHIP PROFILE

The *Havhingsten* is about 30m (98ft) long but only 3.8m (12½ft) wide. This long, narrow shape makes it fast and enables it to sail in water less than 1m (3¼ft) deep.

LONGSHIP STRUCTURE



▲ PARREL

Attached by ropes, a horseshoe-shaped piece of wood known as a parrel held the yard onto the mast. Viking ship ropes were typically made of plant or animal fibres.

► MAST AND MAST FISH

The mast fitted into a socket in a block of wood called the kelson at the bottom of the boat. It was then secured in place by a horizontal piece of timber on the deck known as a mast fish.



◀ CLEAT

The ropes that controlled the sail and the yard (the mast's horizontal wooden spar) were fastened to fixtures called cleats along the hull.

► STRENGTHENING TIMBER

Vertical timbers called top-ribs reinforced the hull's upper structure. They were butted into the gunwale and extended down over the first three strakes, to which they were secured with iron nails.

